

WhatsApp Support Gateway - Video Recording Teleprompter

Silent-recording pacing guide with exact hold times per action for post-recording voiceover sync.

RECORDING STRATEGY & ARCHITECTURAL HIGHLIGHTS:

- **Phone-Based Identity:** Meta webhooks pass authenticated customer phone numbers, enabling zero-friction lookups without login.
- **Simulated Phone Column:** Point out the header selector (Single Order: 15551234567, Multi-Package: 15557778899, New Customer: 15550000000).
- **Telemetry Cockpit:** Walk through sub-20ms Latency, HMAC Security, Intent Router, Session State, and Event Stream.
- **All 4 Order Cases:** Single order auto-resolution, multi-package disambiguation, new customer fallback, and cross-phone Anti-IDOR privacy shield.
- **Hold Time:** Perform each action, then pause on screen for the indicated seconds before clicking next!
- Total video runtime: **~2 minutes 36 seconds** (156 seconds).

STEP 0: INTRO, PHONE IDENTITY & TELEMETRY COCKPIT [■■ HOLD ON SCREEN: 16s]

■ ACTION:

Hover mouse over WhatsApp phone on left, point to Simulated Phone in header, then hover over the Telemetry Cockpit on right.

■ ON SCREEN:

Browser at localhost:8000/demo. Left: WhatsApp simulator. Top: Phone presets. Right: Latency (18ms), HMAC (VALID), Intent (MENU), Session State (ACTIVE_BOT), and real-time backend event stream.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 16s):

"Welcome to our WhatsApp Support Gateway demo. Built with FastAPI and Python, this enterprise system eliminates login friction by using protocol-verified WhatsApp phone numbers from Meta webhooks. On the right, our live telemetry cockpit monitors sub-20ms latency, HMAC-SHA256 signature verification, deterministic intent routing, and real-time backend event logs."

STEP 1: SINGLE ORDER AUTO-RESOLUTION (ZERO TYPING) [■■ HOLD ON SCREEN: 12s]

■ ACTION:

With simulated phone 15551234567, click the native button in phone: ■ Track Order.

■ ON SCREEN:

Inbound webhook logged. Message ticks turn blue (read receipt), 3 typing dots appear, and bot immediately resolves #ORD-1001 (FedEx, in-transit) without typing an order number.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 12s):

"Because Meta passes the customer's verified phone number, our gateway checks the database automatically. Notice the sub-20ms latency, blue read receipt, and typing indicator: the customer never had to type an order number—their shipment details are on screen in one tap."

STEP 2: CONVERSATIONAL MEMORY & RETURN POLICY [■■ HOLD ON SCREEN: 10s]

■ ACTION:

Click chip: 2. Can I return it? (or type it in the phone input).

■ ON SCREEN:

Bot resolves pronoun 'it' to #ORD-1001. Because status is 'SHIPPED', the bot strictly enforces return policy: returns require delivery first.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 10s):

"Notice conversational context: asking 'Can I return it?' resolves 'it' to our active order. Because the package is still in transit, the bot enforces store policy and explains that returns require delivery first."

STEP 3: MULTI-PACKAGE CUSTOMER (DISAMBIGUATION) [■■ HOLD ON SCREEN: 14s]

■ ACTION:

Click purple chip: ■ Demo: Multi-Package
(or switch header dropdown to *Multi-Package: 15557778899* and tap [■ Track Order]).

■ ON SCREEN:

Bot detects 2 active packages for phone +1 555-777-8899 (#ORD-1004 Smart Fitness Watch & #ORD-1005 Soundbar). Displays native buttons: [■ #ORD-1004] and [■ #ORD-1005].

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 14s):

"What if a customer ordered multiple packages? Switching to a customer with multiple active orders, the gateway automatically detects both packages and presents native interactive buttons to disambiguate with zero confusion."

STEP 4: 1-TAP PACKAGE SELECTION & LIVE TRACKING [■■ HOLD ON SCREEN: 10s]

■ ACTION:

Tap the interactive button: ■ #ORD-1004.

■ ON SCREEN:

Bot instantly retrieves and displays the DHL Express tracking details and ETA for the Smart Fitness Watch.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 10s):

"Tapping the package button instantly pulls up the DHL Express tracking status for that exact shipment, with follow-up action buttons attached."

STEP 5: NEW CUSTOMER SCENARIO (0 ORDERS ON FILE) [■■ HOLD ON SCREEN: 10s]

■ ACTION:

In header dropdown, select: New Customer (0 Orders) (15550000000), click ■ Reset Session, then tap ■ Track Order.

■ ON SCREEN:

Bot replies: 'We could not find any active orders associated with your WhatsApp number. If you placed your order under a different number or email, please reply with your order number.'

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 10s):

"If a new customer or unknown number reaches out, the gateway gracefully detects zero orders on file and prompts them for an order number in case they purchased as a gift or under an email."

STEP 6: CROSS-PHONE ORDER & ANTI-IDOR PRIVACY SHIELD [■■ HOLD ON SCREEN: 12s]

■ ACTION:

In header dropdown, switch back to: Single Order (ORD-1001).
Click chip: 3. Track #ORD-1002 (Security Check)

■ ON SCREEN:

Bot intercepts query for #ORD-1002 (registered to another phone 15559876543). Anti-IDOR security shield engages, blocks shipment details, and prompts for the last 4 digits on file.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 12s):

"To prevent privacy leaks and IDOR attacks, if a customer queries an order belonging to a different phone number, our security shield intercepts the request and demands the last 4 digits of the phone number on file."

STEP 7: PIN AUTHENTICATION UNLOCKS ORDER [■■ HOLD ON SCREEN: 8s]

■ ACTION:

Click chip: 4. Verify: 6543 (or type 6543).

■ ON SCREEN:

Bot replies: '■ Security Verification Successful!' and unlocks #ORD-1002 status (PROCESSING).

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 8s):

"Entering the 4 digits validates ownership, unlocks the session, and securely presents the order status."

STEP 8: DELIVERED ORDER RETURN & PREPAID PDF LABEL [■■ HOLD ON SCREEN: 15s]

■ ACTION:

Click chip 5. Return #ORD-1003, then chip 6. Verify: 2233 (Get PDF).
Then click the PDF Document Card in phone.

■ ON SCREEN:

Bot verifies delivered order #ORD-1003, approves return, and attaches downloadable PDF card. Clicking opens vector PDF return label in new tab with Code128 barcode.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 15s):

"For delivered orders, the bot evaluates return eligibility and programmatically generates a vector PDF shipping label using ReportLab, complete with prepaid routing, a Code128 barcode, and an RMA packing slip."

STEP 9: SHOPIFY ADMIN API ADAPTER SYNC [■■ HOLD ON SCREEN: 11s]

■ ACTION:

(Return to demo tab) Click chip: 7. Shopify #1001

■ ON SCREEN:

Telemetry shows routing via ShopifyOrderAdapter. Bot returns live line items and fulfillment status.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 11s):

"Using hexagonal architecture, a single configuration switch connects our gateway to the live Shopify Admin REST API, parsing real-time line items and fulfillment stages seamlessly."

STEP 10: HUMAN ESCALATION & STRICT BOT MUTING [■■ HOLD ON SCREEN: 12s]

■ ACTION:

Click chip: 8. Talk to Human (or type human).

■ ON SCREEN:

Session State turns amber **ESCALATED_HUMAN**. Telemetry logs external webhook to Zendesk/n8n. Operator Desk opens live 2-way WebSocket bridge.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 12s):

"When a customer needs human assistance, an external webhook fires to Zendesk, the bot mutes itself immediately, and our Operator Station connects via a real-time WebSocket bridge."

STEP 11: ANTI-SPAM VERIFICATION & LIVE 2-WAY CHAT [■■ HOLD ON SCREEN: 14s]

■ ACTION:

In phone chat input, type: Where is my refund? and send. (Bot stays silent).

In Operator Desk on right, type: Hi, Agent Sarah here! Your refund of \$189.50 has been released. and click Send as Agent.

■ ON SCREEN:

Bot does NOT reply (zero spam). Customer message streams live to Operator Desk. Agent's reply dispatches directly to phone with blue ■ Agent Sarah badge.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 14s):

"Customer follow-ups never trigger annoying bot spam—the bot stays muted while messages stream live to the operator. The agent replies directly from the console, appearing instantly on the customer's WhatsApp."

STEP 12: SESSION RESOLUTION [■■ HOLD ON SCREEN: 7s]

■ ACTION:

Click button: Unmute Bot / Resolve

■ ON SCREEN:

Operator Station closes. Session State returns to green **ACTIVE_BOT**.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 7s):

"Once resolved, clicking 'Unmute Bot' re-enables automated bot handling for future customer inquiries."

STEP 13: PRODUCTION RIGOR & 103 AUTOMATED TESTS [■■ HOLD ON SCREEN: 12s]

■ ACTION:

Switch to Terminal window and run: `uv run pytest`

■ ON SCREEN:

All 103 tests pass green in under 1.6 seconds.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 12s):

"Under the hood, the system is backed by Docker Compose, Redis Streams for horizontal scaling, strict mypy typing, and 103 automated tests passing in under two seconds."

Master Quick-Reference Cheat Sheet (With Wait Times)

Action (Click / Type)	■ ■ Hold Time	What Happens on Screen	Voiceover Script (To Read Aloud)
Hover mouse over WhatsApp phone on left, point to Simulated Phone in header, then hover over the Telemetry Cockpit on right.	16s	Browser at localhost:8000/demo. Left: WhatsApp simulator. Top: Phone presets. Right: Latency (18ms), HMAC (VALID), Intent (MENU), Session State (ACTIVE_BOT), and real-time backend event stream.	"Welcome to our WhatsApp Support Gateway demo. Built with FastAPI and Python, this enterprise system eliminates login friction by using protocol-verified WhatsApp phone numbers from Meta webhooks. On the right, our live telemetry cockpit monitors sub-20ms latency, HMAC-SHA256 signature verification, deterministic intent routing, and real-time backend event logs."
With simulated phone 15551234567, click the native button in phone: ■ Track Order.	12s	Inbound webhook logged. Message ticks turn blue (read receipt), 3 typing dots appear, and bot immediately resolves #ORD-1001 (FedEx, in-transit) without typing an order number.	"Because Meta passes the customer's verified phone number, our gateway checks the database automatically. Notice the sub-20ms latency, blue read receipt, and typing indicator: the customer never had to type an order number—their shipment details are on screen in one tap."
Click chip: 2. Can I return it? (or type it in the phone input).	10s	Bot resolves pronoun 'it' to #ORD-1001. Because status is 'SHIPPED', the bot strictly enforces return policy: returns require delivery first.	"Notice conversational context: asking 'Can I return it?' resolves 'it' to our active order. Because the package is still in transit, the bot enforces store policy and explains that returns require delivery first."
Click purple chip: ■ Demo: Multi-Package (or switch header dropdown to Multi-Package: 15557778899 and tap [■ Track Order]).	14s	Bot detects 2 active packages for phone +1 555-777-8899 (#ORD-1004 Smart Fitness Watch & #ORD-1005 Soundbar). Displays native buttons: [■ #ORD-1004] and [■ #ORD-1005].	"What if a customer ordered multiple packages? Switching to a customer with multiple active orders, the gateway automatically detects both packages and presents native interactive buttons to disambiguate with zero confusion."
Tap the interactive button: ■ #ORD-1004.	10s	Bot instantly retrieves and displays the DHL Express tracking details and ETA for the Smart Fitness Watch.	"Tapping the package button instantly pulls up the DHL Express tracking status for that exact shipment, with follow-up action buttons attached."
In header dropdown, select: New Customer (0 Orders) (15550000000), click ■ Reset Session, then tap ■ Track Order.	10s	Bot replies: 'We could not find any active orders associated with your WhatsApp number. If you placed your order under a different number or email, please reply with your order number.'	"If a new customer or unknown number reaches out, the gateway gracefully detects zero orders on file and prompts them for an order number in case they purchased as a gift or under an email."
In header dropdown, switch back to: Single Order (ORD-1001). Click chip: 3. Track #ORD-1002 (Security Check)	12s	Bot intercepts query for #ORD-1002 (registered to another phone 15559876543). Anti-IDOR security shield engages, blocks shipment details, and prompts for the last 4 digits on file.	"To prevent privacy leaks and IDOR attacks, if a customer queries an order belonging to a different phone number, our security shield intercepts the request and demands the last 4 digits of the phone number on file."
Click chip: 4. Verify: 6543 (or type 6543).	8s	Bot replies: '■ Security Verification Successful!' and unlocks #ORD-1002 status (PROCESSING).	"Entering the 4 digits validates ownership, unlocks the session, and securely presents the order status."
Click chip 5. Return #ORD-1003, then chip 6. Verify: 2233 (Get PDF). Then click the PDF Document Card in phone.	15s	Bot verifies delivered order #ORD-1003, approves return, and attaches downloadable PDF card. Clicking opens vector PDF return label in new tab with Code128 barcode.	"For delivered orders, the bot evaluates return eligibility and programmatically generates a vector PDF shipping label using ReportLab, complete with prepaid routing, a Code128 barcode, and an RMA packing slip."

(Return to demo tab) Click chip: 7. Shopify #1001	11s	Telemetry shows routing via ShopifyOrderAdapter. Bot returns live line items and fulfillment status.	"Using hexagonal architecture, a single configuration switch connects our gateway to the live Shopify Admin REST API, parsing real-time line items and fulfillment stages seamlessly."
Click chip: 8. Talk to Human (or type human).	12s	Session State turns amber ESCALATED_HUMAN. Telemetry logs external webhook to Zendesk/n8n. Operator Desk opens live 2-way WebSocket bridge.	"When a customer needs human assistance, an external webhook fires to Zendesk, the bot mutes itself immediately, and our Operator Station connects via a real-time WebSocket bridge."
In phone chat input, type: Where is my refund? and send. (Bot stays silent). In Operator Desk on right, type: Hi, Agent Sarah here! Your refund of \$189.50 has been released. and click Send as Agent.	14s	Bot does NOT reply (zero spam). Customer message streams live to Operator Desk. Agent's reply dispatches directly to phone with blue ■ Agent Sarah badge.	"Customer follow-ups never trigger annoying bot spam—the bot stays muted while messages stream live to the operator. The agent replies directly from the console, appearing instantly on the customer's WhatsApp."
Click button: Unmute Bot / Resolve	7s	Operator Station closes. Session State returns to green ACTIVE_BOT.	"Once resolved, clicking 'Unmute Bot' re-enables automated bot handling for future customer inquiries."
Switch to Terminal window and run: uv run pytest	12s	All 103 tests pass green in under 1.6 seconds.	"Under the hood, the system is backed by Docker Compose, Redis Streams for horizontal scaling, strict mypy typing, and 103 automated tests passing in under two seconds."